

Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.11)

Metric

$$ds^2 = \left((a_1^2 \rho^{4m} + 2a_1 a_2 \rho^{2m} + a_2^2) \rho^{2m^2-2m} \right) d\rho \otimes d\rho + \left(-(a_1^2 \rho^{4m} + 2a_1 a_2 \rho^{2m} + a_2^2) \rho^{2m^2-2m} \right) dt \otimes dt + \left(\frac{a_2^2 \rho^2 + a_1^2 \rho^{4m+2} + 2a_1 a_2 \rho^{2m+2}}{\rho^{2m}} \right) d\varphi \otimes d\varphi + \left(\frac{\rho^{2m}}{a_1^2 \rho^{4m} + 2a_1 a_2 \rho^{2m} + a_2^2} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (\rho, t, \varphi, z)$$

Metric Functions

$$G = a_1 \rho^m + \frac{a_2}{\rho^m}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} a_1 \rho^{(m^2)} \rho^m + \frac{\sqrt{2} a_2 \rho^{(m^2)}}{2 \rho^m} \right) d\rho + \left(\frac{1}{2} \sqrt{2} a_1 \rho^{(m^2)} \rho^m + \frac{\sqrt{2} a_2 \rho^{(m^2)}}{2 \rho^m} \right) dt$$

$$n = \left(-\frac{1}{2} \sqrt{2} a_1 \rho^{(m^2)} \rho^m - \frac{\sqrt{2} a_2 \rho^{(m^2)}}{2 \rho^m} \right) d\rho + \left(\frac{1}{2} \sqrt{2} a_1 \rho^{(m^2)} \rho^m + \frac{\sqrt{2} a_2 \rho^{(m^2)}}{2 \rho^m} \right) dt$$

$$m = \left(\frac{1}{2} \sqrt{2} a_1 \rho \rho^m + \frac{\sqrt{2} a_2 \rho}{2 \rho^m} \right) d\varphi + \left(\frac{i \sqrt{2}}{2 \left(a_1 \rho^m + \frac{a_2}{\rho^m} \right)} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} a_1 \rho \rho^m + \frac{\sqrt{2} a_2 \rho}{2 \rho^m} \right) d\varphi + \left(-\frac{i \sqrt{2}}{2 \left(a_1 \rho^m + \frac{a_2}{\rho^m} \right)} \right) dz$$

Contravariant components

$$l = \left(\frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \right) d\rho + \left(-\frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \right) dt$$

$$n = \left(-\frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \right) d\rho + \left(-\frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \right) dt$$

$$m = \left(\frac{\rho^m}{\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2 \rho} \right) d\varphi + \left(\frac{i \sqrt{2} a_1 \rho^{2m}}{2 \rho^m} + \frac{i \sqrt{2} a_2}{2 \rho^m} \right) dz$$

$$\bar{m} = \left(\frac{\rho^m}{\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2 \rho} \right) d\varphi + \left(-\frac{i \sqrt{2} a_1 \rho^{2m}}{2 \rho^m} - \frac{i \sqrt{2} a_2}{2 \rho^m} \right) dz$$

Directional Derivatives

$$D = \frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \partial_\rho - \frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \partial_t$$

$$\Delta = -\frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \partial_\rho - \frac{\rho^m}{(\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2) \rho^{(m^2)}} \partial_t$$

$$\delta = \frac{\rho^m}{\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2 \rho} \partial_\varphi + \frac{i \sqrt{2} a_1 \rho^{2m}}{2 \rho^m} + \frac{i \sqrt{2} a_2}{2 \rho^m} \partial_z$$

$$\bar{\delta} = \frac{\rho^m}{\sqrt{2} a_1 \rho^{2m} + \sqrt{2} a_2 \rho} \partial_\varphi - \frac{i \sqrt{2} a_1 \rho^{2m}}{2 \rho^m} - \frac{i \sqrt{2} a_2}{2 \rho^m} \partial_z$$

Spin Coefficients

$$\rho = -\frac{\rho^m}{2(\sqrt{2}a_1\rho\rho^{2m} + \sqrt{2}a_2\rho)\rho^{(m^2)}}$$

$$\sigma = -\frac{\sqrt{2}a_1m\rho^{3m}}{2(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} + \frac{\sqrt{2}a_2m\rho^m}{2(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} - \frac{\sqrt{2}a_1\rho^{3m}}{4(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} - \frac{\sqrt{2}a_2\rho^m}{4(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}}$$

$$\kappa = 0$$

$$\mu = -\frac{\rho^m}{2(\sqrt{2}a_1\rho\rho^{2m} + \sqrt{2}a_2\rho)\rho^{(m^2)}}$$

$$\lambda = -\frac{\sqrt{2}a_1m\rho^{3m}}{2(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} + \frac{\sqrt{2}a_2m\rho^m}{2(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} - \frac{\sqrt{2}a_1\rho^{3m}}{4(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}} - \frac{\sqrt{2}a_2\rho^m}{4(a_1^2\rho\rho^{4m} + 2a_1a_2\rho\rho^{2m} + a_2^2\rho)\rho^{(m^2)}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{a_1m^2\rho^{3m}}{2(\sqrt{2}a_1^2\rho\rho^{4m} + 2\sqrt{2}a_1a_2\rho\rho^{2m} + \sqrt{2}a_2^2\rho)\rho^{(m^2)}} + \frac{a_2m^2\rho^m}{2(\sqrt{2}a_1^2\rho\rho^{4m} + 2\sqrt{2}a_1a_2\rho\rho^{2m} + \sqrt{2}a_2^2\rho)\rho^{(m^2)}} \\ + \frac{a_1m\rho^{3m}}{2(\sqrt{2}a_1^2\rho\rho^{4m} + 2\sqrt{2}a_1a_2\rho\rho^{2m} + \sqrt{2}a_2^2\rho)\rho^{(m^2)}} - \frac{a_2m\rho^m}{2(\sqrt{2}a_1^2\rho\rho^{4m} + 2\sqrt{2}a_1a_2\rho\rho^{2m} + \sqrt{2}a_2^2\rho)\rho^{(m^2)}}$$

$$\gamma = \frac{a_1 m^2 \rho^{3m}}{2(\sqrt{2}a_1^2 \rho \rho^{4m} + 2\sqrt{2}a_1 a_2 \rho \rho^{2m} + \sqrt{2}a_2^2 \rho) \rho^{(m^2)}} + \frac{a_2 m^2 \rho^m}{2(\sqrt{2}a_1^2 \rho \rho^{4m} + 2\sqrt{2}a_1 a_2 \rho \rho^{2m} + \sqrt{2}a_2^2 \rho) \rho^{(m^2)}} \\ + \frac{a_1 m \rho^{3m}}{2(\sqrt{2}a_1^2 \rho \rho^{4m} + 2\sqrt{2}a_1 a_2 \rho \rho^{2m} + \sqrt{2}a_2^2 \rho) \rho^{(m^2)}} - \frac{a_2 m \rho^m}{2(\sqrt{2}a_1^2 \rho \rho^{4m} + 2\sqrt{2}a_1 a_2 \rho \rho^{2m} + \sqrt{2}a_2^2 \rho) \rho^{(m^2)}}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{2a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{2a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{2a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{2a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\begin{aligned} \Psi_0 = & \frac{a_1^2 m^3 \rho^{6m}}{(a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} - \frac{a_2^2 m^3 \rho^{2m}}{(a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & + \frac{3 a_1^2 m^2 \rho^{6m}}{2 (a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & - \frac{3 a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} + \frac{3 a_2^2 m^2 \rho^{2m}}{2 (a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & + \frac{a_1^2 m \rho^{6m}}{2 (a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} - \frac{a_2^2 m \rho^{2m}}{2 (a_1^4 \rho^2 \rho^{8m} + 4 a_1^3 a_2 \rho^2 \rho^{6m} + 6 a_1^2 a_2^2 \rho^2 \rho^{4m} + 4 a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \end{aligned}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{a_1^2 m^2 \rho^{6m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} + \frac{a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ - \frac{a_2^2 m^2 \rho^{2m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} - \frac{a_2^2 m \rho^{6m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ + \frac{a_2^2 m \rho^{2m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}}$$

$$\Psi_3 = 0$$

$$\begin{aligned} \Psi_4 = & \frac{a_1^2 m^3 \rho^{6m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} - \frac{a_2^2 m^3 \rho^{2m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & + \frac{3a_1^2 m^2 \rho^{6m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & - \frac{3a_1 a_2 m^2 \rho^{4m}}{(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} + \frac{3a_2^2 m^2 \rho^{2m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \\ & + \frac{a_1^2 m \rho^{6m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} - \frac{a_2^2 m \rho^{2m}}{2(a_1^4 \rho^2 \rho^{8m} + 4a_1^3 a_2 \rho^2 \rho^{6m} + 6a_1^2 a_2^2 \rho^2 \rho^{4m} + 4a_1 a_2^3 \rho^2 \rho^{2m} + a_2^4 \rho^2) \rho^{2m^2}} \end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"